

Stat 145 (Multivariate Statistics)

Lesson 1.3 Marginal and Conditional Distributions of Multivariate Normal Random Vectors

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Learning Outcomes

1. Derive additional properties about the multivariate normal distributions.
2. Derive the marginal and conditional distributions of multivariate normal random vectors.

Result 1: All subsets of $\mathbf{Y} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are normally distributed.

If we respectively partition $\mathbf{Y}$, its mean vector $\boldsymbol{\mu}$ and its covariance matrix $\boldsymbol{\Sigma}$ as

$$\mathbf{Y}_{p \times 1} = \begin{bmatrix} \mathbf{Y}_1 \\ - \frac{q \times 1}{(p-q) \times 1} - \end{bmatrix}$$
$$\boldsymbol{\mu}_{p \times 1} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ - \frac{q \times 1}{(p-q) \times 1} - \end{bmatrix}$$

and

$$\boldsymbol{\Sigma}_{p \times p} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & | & \boldsymbol{\Sigma}_{12} \\ - \frac{q \times q}{(p-q) \times q} - & | & - \frac{q \times (p-q)}{(p-q) \times (p-q)} - \end{bmatrix}$$

then $\mathbf{Y}_1$ is distributed as $N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$. We call this the marginal distribution of $\mathbf{Y}_1$.

Example 1.3.1

If $\mathbf{Y}$ is distributed as $N_5(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, find the distribution of $\begin{bmatrix} X_2 \\ X_4 \end{bmatrix}$.

Solution

We can set

$$\mathbf{X}_1 = \begin{bmatrix} X_2 \\ X_4 \end{bmatrix},$$

$$\boldsymbol{\mu}_1 = \begin{bmatrix} \mu_2 \\ \mu_4 \end{bmatrix},$$

and

$$\boldsymbol{\Sigma}_{11} = \begin{bmatrix} \sigma_{22} & \sigma_{24} \\ \sigma_{42} & \sigma_{44} \end{bmatrix}.$$

With this assignment we can rearranged the random vector $\mathbf{X}$, the mean vector $\boldsymbol{\mu}$, and the covariance matrix $\boldsymbol{\Sigma}$ as follows

$$\mathbf{X} = \begin{bmatrix} X_2 \\ X_4 \\ \bar{X}_1 \\ X_3 \\ X_5 \end{bmatrix},$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_2 \\ \mu_4 \\ \mu_1 \\ \mu_3 \\ \mu_5 \end{bmatrix},$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{22} & \sigma_{24} & | & \sigma_{12} & \sigma_{23} & \sigma_{25} \\ \sigma_{42} & \sigma_{44} & | & \sigma_{14} & \sigma_{34} & \sigma_{45} \\ \hline \sigma_{12} & \sigma_{14} & | & \sigma_{11} & \sigma_{13} & \sigma_{15} \\ \sigma_{23} & \sigma_{34} & | & \sigma_{13} & \sigma_{33} & \sigma_{35} \\ \sigma_{25} & \sigma_{45} & | & \sigma_{15} & \sigma_{35} & \sigma_{55} \end{bmatrix}.$$

Therefore, from **Result 1**, $\mathbf{X}_1 = \begin{bmatrix} X_2 \\ X_4 \end{bmatrix}$ is distributed as $N_2(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$, where

$$\boldsymbol{\mu}_1 = \begin{bmatrix} \mu_2 \\ \mu_4 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma}_{11} = \begin{bmatrix} \sigma_{22} & \sigma_{24} \\ \sigma_{42} & \sigma_{44} \end{bmatrix}$$

Result 2:

a) If $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent, then $Cov(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{0}$ a $q_1 \times q_2$ matrix of zeros.

b) If $\begin{bmatrix} \mathbf{Y}_1 \\ \bar{\mathbf{Y}}_2 \end{bmatrix}$ is distributed as $N_{q_1+q_2} \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & | & \boldsymbol{\Sigma}_{12} \\ \hline \bar{\boldsymbol{\Sigma}}_{21} & | & \bar{\boldsymbol{\Sigma}}_{22} \end{bmatrix} \right)$,

then $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent if and only if $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

c) If $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent and are distributed as $N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$, respectively, then $\begin{bmatrix} \mathbf{Y}_1 \\ \bar{\mathbf{Y}}_2 \end{bmatrix}$ is distributed as

$$N_{q_1+q_2} \left(\begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{11} & | & \mathbf{0} \\ \hline \mathbf{0}^T & | & \bar{\boldsymbol{\Sigma}}_{22} \end{bmatrix} \right)$$

Example 1.3.2

Let $\mathbf{Y}_{3 \times 1}$ be distributed as $N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Are Y_1 and Y_2 independent? What about (Y_1, Y_2) and Y_3 ?

Solution:

Looking at the variance-covariance matrix $\sigma_{12} = 1$, hence, Y_1 and Y_2 are not independent. To answer the second question we look at the following partitioning of $\boldsymbol{\Sigma}$

$$\boldsymbol{\Sigma} = \left[\begin{array}{cc|c} 4 & 1 & 0 \\ 1 & 3 & 0 \\ \hline 0 & 0 & 2 \end{array} \right].$$

Here,

$$\boldsymbol{\Sigma}_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus, (Y_1, Y_2) is independent of Y_3 . This implies Y_3 is independent of Y_1 and Y_2 .

Result 3

Let $\mathbf{Y} = \begin{bmatrix} \mathbf{Y}_1 \\ \mathbf{Y}_2 \end{bmatrix}$ be distributed as $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix},$$

$$\boldsymbol{\Sigma} = \left[\begin{array}{cc|c} \boldsymbol{\Sigma}_{11} & & \boldsymbol{\Sigma}_{12} \\ & \boldsymbol{\Sigma}_{21} & \\ \hline & & \boldsymbol{\Sigma}_{22} \end{array} \right],$$

and

$$|\boldsymbol{\Sigma}_{22}| > 0.$$

Then, the conditional distribution of $\mathbf{Y}_1$, given that $\mathbf{Y}_2 = \mathbf{y}_2$, is normal with mean and covariance, respectively, given by

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{y}_2 - \boldsymbol{\mu}_2)$$

and

$$\boldsymbol{\Sigma}_{1|2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$$

Example 1.3.3

Consider

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix} \sim N \left(\begin{bmatrix} 1 \\ 2 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 & 0 & 1 & 3 \\ 0 & 4 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 3 & 1 & 1 & 9 \end{bmatrix} \right).$$

Find the conditional distribution of $\begin{bmatrix} Y_1 \\ Y_3 \end{bmatrix}$ given $\begin{bmatrix} Y_2 \\ Y_4 \end{bmatrix}$.

Solution: Left as a classroom exercise!

The random vector can be re-arranged as

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_3 \\ \bar{Y}_2 \\ Y_4 \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_1 \\ \bar{\mathbf{Y}}_2 \end{bmatrix}$$

Similarly, we can re-arrange the mean vector and the variance-covariance matrix as follows:

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_3 \\ \mu_2 \\ \mu_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 3 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 1 & 0 & 3 \\ 1 & 3 & 1 & 1 \\ 0 & 1 & 4 & 1 \\ 3 & 1 & 1 & 9 \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}$$

Therefore, according to **Result 3**, the conditional distribution of $\begin{bmatrix} Y_1 \\ Y_3 \end{bmatrix}$ given $\begin{bmatrix} Y_2 \\ Y_4 \end{bmatrix}$ is normal with

$$\begin{aligned} \boldsymbol{\mu}_{1|2} &= \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{y}_2 - \boldsymbol{\mu}_2) \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 1 & 9 \end{bmatrix}^{-1} \left(\begin{bmatrix} y_2 \\ y_4 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 3 \\ 1 & 1 \end{bmatrix} \frac{1}{35} \begin{bmatrix} 9 & -1 \\ -1 & 4 \end{bmatrix} \left(\begin{bmatrix} y_2 - 2 \\ y_4 - 3 \end{bmatrix} \right) \\ &= \frac{1}{35} \begin{bmatrix} -3y_2 + 12y_4 + 5 \\ 8y_2 + 3y_4 - 25 \end{bmatrix} \end{aligned}$$

and

$$\begin{aligned} \boldsymbol{\Sigma}_{1|2} &= \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} \\ &= \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 1 & 9 \end{bmatrix}^{-1} \begin{bmatrix} 0 & 1 \\ 3 & 1 \end{bmatrix} \\ &= \frac{1}{35} \begin{bmatrix} 104 & 26 \\ 26 & 94 \end{bmatrix} \end{aligned}$$